

Bayesian Poisson Factorization for Recommender Systems

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Machine Learning II Oral Exam
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• Core Papers

- *Scalable Recommendation with Hierarchical Poisson Factorization* (Gopalan, Hofman, Blei)
- *Bayesian Nonparametric Poisson Factorization for Recommendation Systems* (Gopalan, Ruiz, Ranganath, Blei)

• Brief Introduction

- Proposes Hierarchical Poisson Factorization for implicit-feedback recommendation and a scalable variational inference algorithm
- Extends to a Bayesian nonparametric model with adaptive latent dimensionality and comparable computational complexity

• Basic Idea

- Recommender systems match user-item interaction by users' preferences and items' attributes
- BPF: Poisson likelihood on feedback & Gamma prior on parameter

1.1 HPF: Model

1. For each user u :

- (a) Sample activity $\xi_u \sim \text{Gamma}(a', a'/b')$.
- (b) For each component k , sample preference

$$\theta_{uk} \sim \text{Gamma}(a, \xi_u).$$

2. For each item i :

- (a) Sample popularity $\eta_i \sim \text{Gamma}(c', c'/d')$.
- (b) For each component k , sample attribute

$$\beta_{ik} \sim \text{Gamma}(c, \eta_i).$$

3. For each user u and item i , sample rating

$$y_{ui} \sim \text{Poisson}(\theta_u^\top \beta_i).$$

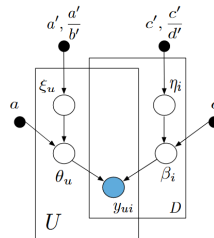


Figure 1: The hierarchical Poisson factorization model.

(a) Hierarchical Poisson Factorization

(b) Hierarchical Gamma Priors

- Infer posterior of latent user preferences θ_u and item attributes β_i from user-item interaction data
- Recommend items by ranking $\mathbb{P}(u)$ using

$$\text{score}_{ui} = \mathbb{E}[\theta_u^\top \beta_i \mid y]$$

1.2 HPF: Properties

- **Sparse factors & long-tail behavior:** Gamma priors capture the long-tail nature of sparse consumption data.

$$p(x) \propto x^{\alpha-1} e^{-\beta x},$$

- **Zeros are downweighted:** HPF downweights the items that each user did not consume.
- **Fast inference on sparse data**

$$p(y_{ui} \mid \theta_u, \beta_i) = \frac{(\theta_u^\top \beta_i)^{y_{ui}} \exp\{-\theta_u^\top \beta_i\}}{y_{ui}!}$$

$$\log p(y \mid \theta, \beta) = \sum_{\{u,i: y_{ui}>0\}} \left[y_{ui} \log(\theta_u^\top \beta_i) - \log(y_{ui}!) \right] - \left(\sum_u \theta_u \right)^\top \left(\sum_i \beta_i \right)$$

1.3 HPF: Inference with Variational Methods

Goal. Given observed user-item data $y = \{y_{ui}\}$, infer the latent structure $p(\theta, \beta, \xi, \eta \mid y)$, where θ_{uk} = user preference, β_{ik} = item attribute, ξ_u = user activity, η_i = item popularity.

Workflow. (1) Specify the variational family (mean-field) for $(\theta, \beta, \xi, \eta)$.
(2) Optimize ν via coordinate ascent updates $\Rightarrow$ scalable posterior approximation.

Turn inference into optimization. Variational methods posit a *family* $\mathcal{Q}$ over hidden variables, indexed by free *variational parameters* ν , and solve

$$q^* = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(\theta, \beta, \xi, \eta, z) \parallel p(\theta, \beta, \xi, \eta, z \mid y)).$$

KL divergence measures how far an approximation q is from the true posterior:

$$\text{KL}(q \parallel p) = \mathbb{E}_q \left[\log \frac{q}{p} \right] \geq 0, \quad \text{KL}(q \parallel p) = 0 \Leftrightarrow q = p.$$

Hyperparameter. For a $\text{Gamma}(\alpha, \beta)$ prior,

$$p(x) \propto x^{\alpha-1} e^{-\beta x}.$$

$a', a, c, c' = 0.3$ places substantial mass near zero.

$b' = d' = 1$ normalizes the scale of the prior (mean = $\alpha/\beta = \alpha$)

1.3 HPF: Inference with Variational Methods

$$q(\beta, \theta, \xi, \eta, z) = \prod_{i,k} q(\beta_{ik} | \lambda_{ik}) \prod_{u,k} q(\theta_{uk} | \gamma_{uk}) \prod_u q(\xi_u | \kappa_u) \prod_i q(\eta_i | \tau_i) \prod_{u,i} q(z_{ui} | \phi_{ui}).$$

Initialize user params $\{\gamma_{uk}, \kappa_u^{\text{rte}}\}$ and item params $\{\lambda_{ik}, \tau_i^{\text{rte}}\}$ to the priors with small random offsets. Set activity/popularity shapes:

$$\kappa_u^{\text{shp}} = a' + Ka, \quad \tau_i^{\text{shp}} = c' + Kc.$$

(1) Update multinomials (only for nonzeros): for each (u, i) with $y_{ui} > 0$,

$$\phi_{ui,k} \propto \exp \left\{ \Psi(\gamma_{uk}^{\text{shp}}) - \log \gamma_{uk}^{\text{rte}} + \Psi(\lambda_{ik}^{\text{shp}}) - \log \lambda_{ik}^{\text{rte}} \right\}.$$

(2) User updates: for each user u and component k ,

$$\gamma_{uk}^{\text{shp}} = a + \sum_i y_{ui} \phi_{ui,k}, \quad \gamma_{uk}^{\text{rte}} = \frac{\kappa_u^{\text{shp}}}{\kappa_u^{\text{rte}}} + \sum_i \frac{\lambda_{ik}^{\text{shp}}}{\lambda_{ik}^{\text{rte}}}, \quad \kappa_u^{\text{rte}} = \frac{a'}{b'} + \sum_k \frac{\gamma_{uk}^{\text{shp}}}{\gamma_{uk}^{\text{rte}}}.$$

(3) Item updates: for each item i and component k ,

$$\lambda_{ik}^{\text{shp}} = c + \sum_u y_{ui} \phi_{ui,k}, \quad \lambda_{ik}^{\text{rte}} = \frac{\tau_i^{\text{shp}}}{\tau_i^{\text{rte}}} + \sum_u \frac{\gamma_{uk}^{\text{shp}}}{\gamma_{uk}^{\text{rte}}}, \quad \tau_i^{\text{rte}} = \frac{c'}{d'} + \sum_k \frac{\lambda_{ik}^{\text{shp}}}{\lambda_{ik}^{\text{rte}}}.$$

Stopping rule. Stop if the relative change in validation log-likelihood is below

$$\frac{|\mathcal{L}_{\text{val}}^{(t)} - \mathcal{L}_{\text{val}}^{(t-1)}|}{|\mathcal{L}_{\text{val}}^{(t-1)}|} < 10^{-6}$$

HPF: Derivation

Auxiliary-variable construction. For each user-item pair (u, i) , introduce latent counts

$$z_{uik} \sim \text{Poisson}(\theta_{uk}\beta_{ik}), \quad y_{ui} = \sum_{k=1}^K z_{uik}.$$

Complete conditional of z_{ui} . Independent Poisson variables conditioned on their sum follow a multinomial,

$$z_{ui} \mid \theta, \beta, y \sim \text{Mult}(y_{ui}, \pi_{ui}), \quad \pi_{ui,k} = \frac{\theta_{uk}\beta_{ik}}{\sum_{\ell=1}^K \theta_{u\ell}\beta_{i\ell}}.$$

Coordinate ascent variational update. The closed-form solution:

$$\log q^*(z_{ui}) = \mathbb{E}_{q(\theta, \beta)}[\log p(z_{ui} \mid \theta, \beta, y)] + \text{const.}$$

Using $\log \pi_{ui,k} = \log \theta_{uk} + \log \beta_{ik} - \log(\sum_{\ell} \theta_{u\ell}\beta_{i\ell})$ and noting that the last term is constant in k , we obtain

$$q(z_{ui}) = \text{Mult}(y_{ui}, \phi_{ui}), \quad \phi_{ui,k} \propto \exp(\mathbb{E}_q[\log \theta_{uk}] + \mathbb{E}_q[\log \beta_{ik}]).$$

Under the mean-field assumption,

$$q(\theta_{uk}) = \text{Gamma}(\gamma_{uk}^{\text{shp}}, \gamma_{uk}^{\text{rte}}), \quad q(\beta_{ik}) = \text{Gamma}(\lambda_{ik}^{\text{shp}}, \lambda_{ik}^{\text{rte}}),$$

For $X \sim \text{Gamma}(\alpha, \beta)$,

$$\mathbb{E}[\log X] = \Psi(\alpha) - \log \beta.$$

HPF: Derivation

Complete conditional of θ_{uk} . In the augmented model,

$$z_{uik} \sim \text{Poisson}(\theta_{uk}\beta_{ik}), \quad \theta_{uk} \mid \xi_u \sim \text{Gamma}(a, \xi_u).$$

Collecting all terms involving θ_{uk} in the log joint:

$$\log p(\theta_{uk} \mid \beta, \xi, z) = \left(a - 1 + \sum_i z_{uik}\right) \log \theta_{uk} - \left(\xi_u + \sum_i \beta_{ik}\right) \theta_{uk} + \text{const.}$$

Thus the complete conditional is $\theta_{uk} \mid \beta, \xi, z \sim \text{Gamma}(a + \sum_i z_{uik}, \xi_u + \sum_i \beta_{ik})$

Coordinate ascent update. Replacing latent variables by their variational expectations,

$$\mathbb{E}_q[z_{uik}] = y_{ui}\phi_{ui,k}, \quad \mathbb{E}_q[\beta_{ik}] = \frac{\lambda_{ik}^{\text{shp}}}{\lambda_{ik}^{\text{rte}}}, \quad \mathbb{E}_q[\xi_u] = \frac{\kappa_u^{\text{shp}}}{\kappa_u^{\text{rte}}},$$

$$\text{yields } \gamma_{uk}^{\text{shp}} = a + \sum_i y_{ui}\phi_{ui,k}, \quad \gamma_{uk}^{\text{rte}} = \frac{\kappa_u^{\text{shp}}}{\kappa_u^{\text{rte}}} + \sum_i \frac{\lambda_{ik}^{\text{shp}}}{\lambda_{ik}^{\text{rte}}}$$

Update of the activity parameter ξ_u . From the hierarchical prior,

$$\xi_u \mid \theta \sim \text{Gamma}\left(a' + Ka, \frac{a'}{b'} + \sum_k \theta_{uk}\right),$$

and therefore

$$\kappa_u^{\text{rte}} = \frac{a'}{b'} + \sum_k \mathbb{E}_q[\theta_{uk}] = \frac{a'}{b'} + \sum_k \frac{\gamma_{uk}^{\text{shp}}}{\gamma_{uk}^{\text{rte}}}.$$

2.1 BNP: Model

From PMF to BNP. Traditional matrix factorization fixes the latent dimension K and fails to capture the dimensionality of the data. BNP allows the number of components to be arbitrary.

Key assumptions:

- (i) User weights θ_u and item weights β_i are infinite-dimensional and non-negative.
- (ii) The expected dot product is finite: $\mathbb{E}[\theta_u^\top \beta_i] < \infty$.
- (iii) Item weights β_i are *shared across all users*.

Generative Process:

(1) Infinite item weights: iid Gamma priors.

$$\beta_{ik} \sim \text{Gamma}(a, b), \quad k = 1, 2, \dots, \infty, \quad i = 1, \dots, M.$$

(2) Finite expected dot product via Gamma process for users. To ensure $\mathbb{E}[\theta_u^\top \beta_i] < \infty$, it is sufficient that $\sum_{k=1}^{\infty} \mathbb{E}[\theta_{uk}] < \infty$ (since $\mathbb{E}[\beta_{ik}] < \infty$). We draw θ_u from a Gamma process with finite measure.

2.1 BNP: Model

(3) Sharing components across users via size-biased ordering. Item weights $\beta_i = (\beta_{i1}, \beta_{i2}, \dots)$ are indexed by $k \in \mathbb{N}$, while user weights θ_u from Gamma process draw are unordered. We impose a *size-biased ordering* to align them. Specifically (Zhou & Carin, 2013):

- Draw a global scale $s_u \sim \text{Gamma}(\alpha, c)$.
- Draw stick proportions $v_{uk} \sim \text{Beta}(1, \alpha)$.
- Define

$$\theta_{uk} = s_u v_{uk} \prod_{j < k} (1 - v_{uj}), \quad k = 1, 2, \dots$$

This construction yields a draw from a **Gamma process** with parameters (c, H) , where the total mass of the base measure satisfies $H(\Omega) = \alpha$.

$$\begin{aligned} \beta_{ik} &\sim \text{Gamma}(a, b), \quad i = 1, \dots, M, \quad k = 1, 2, \dots \\ y_{ui} &\sim \text{Poisson} \left(\sum_{k=1}^{\infty} \theta_{uk} \beta_{ik} \right), \quad u = 1, \dots, N, \quad i = 1, \dots, M. \end{aligned}$$

2.2 BNP: Inference with Variational Methods

Joint distribution with auxiliary variables.

$$p(z, \beta, s, v \mid \alpha, c, a, b) = \prod_{u=1}^N p(s_u \mid \alpha, c) \prod_{k=1}^{\infty} \prod_{u=1}^N p(v_{uk} \mid \alpha) \\ \times \prod_{k=1}^{\infty} \prod_{i=1}^M p(\beta_{ik} \mid a, b) \prod_{k=1}^{\infty} \prod_{u=1}^N \prod_{i=1}^M p(z_{ui,k} \mid \theta_{uk}, \beta_{ik}).$$

Mean-field variational family.

$$q(z, \beta, s, v) = \prod_{u=1}^N q(s_u) \prod_{k=1}^{\infty} \prod_{u=1}^N q(v_{uk}) \times \prod_{k=1}^{\infty} \prod_{i=1}^M q(\beta_{ik}) \prod_{u=1}^N \prod_{i=1}^M q(z_{ui}).$$

Variational objective. Typical mean-field methods optimize the KL divergence

$$\text{KL}(q(z, \beta, s, v) \parallel p(z, \beta, s, v \mid y)) = \mathbb{E}_q \left[\log \frac{q}{p} \right],$$

via coordinate ascent, iteratively updating each factor while holding the others fixed.

2.2 BNP: Inference with Variational Methods

Conditional conjugacy. For conditionally conjugate variables, s_u, β_{ik}, z_{ui} , variational distributions are the same as their complete conditionals.

Non-conjugacy of stick proportions. Beta prior on v_{uk} is not conjugate to Poisson likelihood. The complete conditional is a truncated Gamma distribution without closed form updates.

Solution. We therefore use a degenerate delta distribution $q(v_{uk}) = \delta_{\tau_{uk}}(v_{uk})$, truncating at level T :

$$\begin{aligned} q(s_u) &= \text{Gamma}(s_u \mid \gamma_{u,0}, \gamma_{u,1}), \\ q(v_{uk}) &= \begin{cases} \delta_{\tau_{uk}}(v_{uk}), & k \leq T, \\ p(v_{uk}), & k \geq T+1, \end{cases} \\ q(\beta_{ik}) &= \begin{cases} \text{Gamma}(\lambda_{ik,0}, \lambda_{ik,1}), & k \leq T, \\ p(\beta_{ik}), & k \geq T+1, \end{cases} \\ q(z_{ui}) &= \text{Mult}(z_{ui} \mid y_{ui}, \phi_{ui}). \end{aligned}$$

2.2 BNP: Inference with Variational Methods

Initialize. Given truncation level T , for all users u and items i : Initialize user scaling parameters $\{\gamma_{u,0}, \gamma_{u,1}\}$ and item parameters $\{\lambda_{ik,0}, \lambda_{ik,1}\}$ to the prior with a small random offset; Initialize variational parameter τ_{uk} randomly for $k \leq T$.

Repeat until convergence:

1) For each user-item pair (u, i) such that $y_{ui} > 0$:

$$\phi_{ui,k} = \frac{\exp\{R_{ui,k}\}}{\sum_{\ell=1}^T \exp\{R_{ui,\ell}\}}, \quad R_{ui,k} = \mathbb{E}_q[\log \theta_{uk}] + \mathbb{E}_q[\log \beta_{ik}]. \quad (16)$$

2) For each user u :

$$\gamma_{u,0} = \alpha + \sum_{i=1}^M y_{ui}, \quad (9)$$

$$\gamma_{u,1} = c + \mathbb{E}_q \left[\sum_{k=1}^T v_{uk} \left(\prod_{j=1}^{k-1} (1 - v_{uj}) \right) \sum_{i=1}^M \beta_{ik} \right]. \quad (10)$$

For each stick $k \leq T$:

$$\tau_{uk} = \frac{-B_{uk} \pm \sqrt{B_{uk}^2 - 4A_{uk}C_{uk}}}{2A_{uk}}, \quad \text{choose the root in } (0, 1). \quad (12)$$

2.2 BNP: Inference with Variational Methods

3) Item updates. For each item i and each component $k \leq T$:

$$\lambda_{ik,0} = a + \sum_{u=1}^N y_{ui} \phi_{ui,k}, \quad (13)$$

$$\lambda_{ik,1} = b + \sum_{u=1}^N \mathbb{E}_q \left[s_u v_{uk} \prod_{j=1}^{k-1} (1 - v_{uj}) \right]. \quad (14)$$

Output. Variational parameters $\{\gamma, \lambda, \tau, \phi\}$ defining $q(s), q(\beta), q(v), q(z)$.

Computational complexity. For a dataset with N users, M items, and r nonzero observations: $\mathcal{O}(T^2N + TM + Tr)$ The dominant cost is $\mathcal{O}(Tr)$, the complexity of finite Poisson factorization with fixed dimension $K = T$.

Convergence criterion. Iterations stop when the relative change in validation log-likelihood falls below a threshold:

$$\frac{|\mathcal{L}_{\text{val}}^{(t)} - \mathcal{L}_{\text{val}}^{(t-1)}|}{|\mathcal{L}_{\text{val}}^{(t-1)}|} < 10^{-6}.$$

BNP: Derivation

Mean-field CAVI identity. For factor $q(x)$, the coordinate-optimal update is

$$\log q^*(x) = \mathbb{E}_{q(\neg x)}[\log p(y, z, s, v, \beta)] + \text{const.}$$

Poisson additivity. Introduce $z_{ui,k} \sim \text{Pois}(\theta_{uk}\beta_{ik})$ s.t. $y_{ui} = \sum_{k=1}^{\infty} z_{ui,k}$

1) Update for $q(z_{ui})$ Fix (u, i) with $y_{ui} > 0$ and collect $\log p(z_{ui} \mid \theta_u, \beta_i)$:

$$\log p(z_{ui} \mid \theta_u, \beta_i) = \sum_{k \geq 1} \left[z_{ui,k} \log(\theta_{uk}\beta_{ik}) - \log(z_{ui,k}!) \right] + \text{const} \quad \text{s.t.} \quad \sum_k z_{ui,k} = y_{ui}.$$

Thus the complete conditional is multinomial:

$$p(z_{ui} \mid y_{ui}, \theta_u, \beta_i) = \text{Mult}(y_{ui}, \pi_{ui,k} \propto \theta_{uk}\beta_{ik}).$$

Apply CAVI:

$$q^*(z_{ui}) = \text{Mult}(y_{ui}, \phi_{ui}), \quad \phi_{ui,k} \propto \exp \left\{ \mathbb{E}_q[\log \theta_{uk}] + \mathbb{E}_q[\log \beta_{ik}] \right\}.$$

Define $R_{ui,k} = \mathbb{E}_q[\log \theta_{uk}] + \mathbb{E}_q[\log \beta_{ik}]$, then

$$\phi_{ui,k} = \frac{\exp\{R_{ui,k}\}}{\sum_{\ell \geq 1} \exp\{R_{ui,\ell}\}}$$

BNP: Derivation

2) Update for $q(s_u) = \text{Gamma}(\gamma_{u,0}, \gamma_{u,1})$. Write $\theta_{uk} = s_u \pi_{uk}$ with $\pi_{uk} = v_{uk} \prod_{j < k} (1 - v_{uj})$. Collect all ELBO terms involving s_u via the complete conditional:

$$\log p(s_u | \cdot) = (\alpha - 1) \log s_u - c s_u + \sum_{i=1}^M \sum_{k \geq 1} \left[z_{ui,k} \log(s_u \pi_{uk} \beta_{ik}) - s_u \pi_{uk} \beta_{ik} \right] + \text{const.}$$

Keep only $\log s_u$ and s_u terms:

$$\log p(s_u | \cdot) = \left(\alpha - 1 + \sum_i \sum_k z_{ui,k} \right) \log s_u - \left(c + \sum_k \pi_{uk} \sum_i \beta_{ik} \right) s_u + \text{const.}$$

Take $\mathbb{E}_{q(\neg s_u)}[\cdot]$ and use $\sum_k z_{ui,k} = y_{ui}$ and $\mathbb{E}_q[z_{ui,k}] = y_{ui} \phi_{ui,k}$:

$$\log q^*(s_u) = \left(\alpha - 1 + \sum_i y_{ui} \right) \log s_u - \left(c + \mathbb{E}_q \left[\sum_{k \geq 1} \pi_{uk} \sum_i \beta_{ik} \right] \right) s_u + \text{const.}$$

Hence Gamma with

$$\gamma_{u,0} = \alpha + \sum_{i=1}^M y_{ui}, \quad \gamma_{u,1} = c + \mathbb{E}_q \left[\sum_{k \geq 1} v_{uk} \left(\prod_{j < k} (1 - v_{uj}) \right) \sum_{i=1}^M \beta_{ik} \right].$$

2) Stick point update τ_{uk} . With $q(v_{uk}) = \delta_{\tau_{uk}}$, update τ_{uk} by maximizing ELBO:

$$\frac{\partial \mathcal{L}}{\partial \tau_{uk}} = 0 \implies A_{uk}\tau_{uk}^2 + B_{uk}\tau_{uk} + C_{uk} = 0,$$

where (A_{uk}, B_{uk}, C_{uk}) are functions of current sufficient statistics $\{y_{ui}\phi_{ui,\ell}\}$, $\{\mathbb{E}_q[\beta_{i\ell}]\}$ and stick products $\prod_{j<\ell}(1-\tau_{uj})$ (given explicitly in the paper appendix/supplement). Thus

$$\tau_{uk} = \frac{-B_{uk} \pm \sqrt{B_{uk}^2 - 4A_{uk}C_{uk}}}{2A_{uk}}, \quad \text{choose the root in } (0, 1).$$

BNP: Derivation

3) Update for $q(\beta_{ik}) = \text{Gamma}(\lambda_{ik,0}, \lambda_{ik,1})$ (for $k \leq T$). Complete conditional:

$$\log p(\beta_{ik} | \cdot) = (a - 1) \log \beta_{ik} - b\beta_{ik} + \sum_{u=1}^N \left[z_{ui,k} \log(\theta_{uk}\beta_{ik}) - \theta_{uk}\beta_{ik} \right] + \text{const.}$$

Drop terms not depending on β_{ik} :

$$\log p(\beta_{ik} | \cdot) = \left(a - 1 + \sum_u z_{ui,k} \right) \log \beta_{ik} - \left(b + \sum_u \theta_{uk} \right) \beta_{ik} + \text{const.}$$

Take $\mathbb{E}_{q(\cdot)}[\cdot]$:

$$q^*(\beta_{ik}) = \text{Gamma}(\lambda_{ik,0}, \lambda_{ik,1}), \quad \lambda_{ik,0} = a + \sum_{u=1}^N \mathbb{E}_q[z_{ui,k}], \quad \lambda_{ik,1} = b + \sum_{u=1}^N \mathbb{E}_q[\theta_{uk}].$$

Use $\mathbb{E}_q[z_{ui,k}] = y_{ui}\phi_{ui,k}$:

$$\lambda_{ik,0} = a + \sum_{u=1}^N y_{ui}\phi_{ui,k}.$$

And $\theta_{uk} = s_u v_{uk} \prod_{j < k} (1 - v_{uj})$:

$$\lambda_{ik,1} = b + \sum_{u=1}^N \mathbb{E}_q \left[s_u v_{uk} \prod_{j < k} (1 - v_{uj}) \right].$$

• Limitation

- For inference, the variational objective in BNP must be concave in $\tau_{uk} \rightarrow \alpha \geq 1$
- Consequence: the model cannot represent extremely sparse user preferences ($\alpha < 1$).

• Recent Progress in Recommender Systems

- Multimodal inputs & representation learning
- Large Language Models for recommendation

• Downstream Societal Impact

- Preference matching in digital markets
- Political polarization on social media
- Trade-offs between personalization, diversity, and privacy